

4DOTS Landing Page

A fast, dependency-free landing page for 4DOTS, hosted on Cloudflare Pages. Lead submissions are validated by Cloudflare Turnstile and written to a private Google Sheet through a small Google Apps Script web app.

Architecture

```
Visitor
-> Cloudflare Pages static site
-> Google Apps Script web app
    -> Cloudflare Turnstile verification
    -> server-side input validation
    -> private Google Sheet
```

The frontend contains no secret values. The Turnstile secret and Sheet ID stay in Apps Script Properties.

Repository Structure

docs/	Public Cloudflare Pages output
index.html	
styles.css	
app.js	
_headers	
apps-script/	Google Apps Script source
Code.gs	
appsscript.json	
tests/	Automated contract and validation tests
requirements-dev.txt	Pinned local/CI test dependency
wrangler.jsonc	Cloudflare Pages project configuration
assets/	Local design references; not published from docs/

Run Locally

From the repository root:

```
python3 -m http.server 8000 --directory docs
```

Open <http://localhost:8000>. The form stays disabled until its public endpoint and Turnstile site key are configured.

Run Automated Tests

Create the local virtual environment and install the test dependency:

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements-dev.txt
```

Run the suite:

```
.venv/bin/pytest
```

The tests verify the Cloudflare Pages configuration and headers, Apps Script timezone, frontend/backend allowed-value contract, backend validation and formula-injection protection, and JavaScript syntax. GitHub Actions runs the same suite on pushes and pull requests.

Configure the Google Sheet

1. Create a private Google Sheet for leads.
2. Copy the ID from its URL:

```
https://docs.google.com/spreadsheets/d/SHEET_ID/edit
```

3. Do not make the Sheet public. The Apps Script deployment writes to it using the deploying Google account.

The script creates a **Leads** tab and header row automatically on the first valid submission.

Configure Cloudflare Turnstile

1. Create a free Turnstile widget in the Cloudflare dashboard.
2. Add the launch hostname, such as **4dots.pages.dev**, and any custom domain you plan to use.
3. Copy the public **site key** and private **secret key**.
4. Put the site key in **docs/app.js**:

```
turnstileSiteKey: "YOUR_PUBLIC_SITE_KEY",
```

Never put the Turnstile secret key in **docs/** or anywhere else served by Cloudflare Pages.

Deploy Google Apps Script

1. Open [Google Apps Script](#) and create a project.
2. Replace its source with **apps-script/Code.gs**.

3. In **Project Settings**, enable showing the manifest file and replace it with `apps-script/appsscript.json`.
4. Under **Script Properties**, add:

Property	Required	Value
<code>SHEET_ID</code>	Yes	Google Sheet ID
<code>TURNSTILE_SECRET</code>	Yes	Private Turnstile secret key
<code>SHEET_NAME</code>	No	Defaults to <code>Leads</code>
<code>ALLOWED_HOSTNAMES</code>	Recommended	Comma-separated hosts, such as <code>4dots.pages.dev, www.example.com</code>
<code>RETURN_URL</code>	Recommended	Full public landing-page URL

5. Select **Deploy > New deployment > Web app**.
6. Set **Execute as** to yourself.
7. Set access to **Anyone**.
8. Authorize the requested Google Sheets and external-request scopes.
9. Copy the deployed `/exec` URL.
10. Put that public URL in `docs/app.js`:

```
formEndpoint: "YOUR_GOOGLE_APPS_SCRIPT_EXEC_URL",
```

The `/exec` URL is a public endpoint and is safe to place in the frontend. The receiver still requires a valid, single-use Turnstile token and validates all submitted values server-side.

After changing `Code.gs`, create a new Apps Script deployment version so the public endpoint receives the update.

Deploy Cloudflare Pages

1. Push the repository to GitHub.
2. In the Cloudflare dashboard, open **Workers & Pages** and create a Pages application using **Import an existing Git repository**.
3. Connect this repository and use `main` as the production branch.
4. Select no framework preset, leave the build command blank, and use `docs` as the build output directory.
5. Deploy the project. Cloudflare will publish future pushes automatically and create preview deployments for non-production branches.

The checked-in `wrangler.jsonc` keeps the Pages project name, output directory, and compatibility date in source control. Cloudflare will publish the site at:

```
https://4dots.pages.dev/
```

The exact `pages.dev` hostname may differ if the project name is unavailable. After the first deployment, add the final hostname to Turnstile and the Apps Script `ALLOWED_HOSTNAMES` property, then set `RETURN_URL` to the full site URL.

For an authenticated manual deployment with Wrangler:

```
npx wrangler pages deploy docs
```

Security Notes

- Never commit API secrets, Sheet credentials, or Turnstile secrets.
- Keep the Google Sheet private and limit its editors.
- Use `ALLOWED_HOSTNAMES` in Apps Script for production.
- The receiver verifies Turnstile, validates allowed choices, limits field lengths, rejects malformed fields, uses a honeypot, locks concurrent writes, and neutralizes spreadsheet formula injection.
- Cloudflare Pages is appropriate for public lead forms, not passwords, card details, or other sensitive transactions.

Launch Checklist

- Replace both placeholders in `docs/app.js`.
- Submit one valid test lead and confirm it appears in the private Sheet.
- Test invalid email, invalid phone, missing consent, and no selected service.
- Test on mobile and desktop.
- Set `RETURN_URL` and `ALLOWED_HOSTNAMES` to the final Cloudflare Pages or custom-domain hostname.
- Confirm Cloudflare Pages serves the security headers from `docs/_headers`.
- Add a privacy policy before running paid acquisition campaigns.

License

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